

CSCI E-106: Assignment 5 solutions

Due Date: March 6, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

Refer to the CDI data set. The number of active physicians (Y) is to be regressed against total population, total personal income, land area and geographic region ("Geographic region classification is that used by the U.S. Bureau of the Census, where: 1 = NE, 2 = NC, S = 3, W=4")

Data Preparations

```
PMQ1<- read.csv("CDI Data.csv")
# Development Data
Y<-PMQ1$Number.of.active.physicians
X1<-PMQ1$Total.population
X2<-PMQ1$Total.personal.income
X3<-PMQ1$Land.area
X4<-PMQ1$Geographic.region

data<-cbind(Y,X1,X2,X3,X4)
data<-data.frame(data)
```

##a-) Create Dummy variables for Geographic Region.

Conclusion We selected to drop the level with less observations to create the dummies to model the regional categorical variable. See code below.

```
# Check X4 Levels, need 3 dummies, we decided to drop the level with less
table(X4)
```

```
## X4
##   1   2   3   4
## 103 108 152  77

data$X4.1=I(data$X4==1)*1
data$X4.2=I(data$X4==2)*1
data$X4.3=I(data$X4==3)*1
# Drop un-necessary column
data <- data %>% select(-c(X4))
```

##b-) Fit a regression model by using all variables above. Comment on your model.

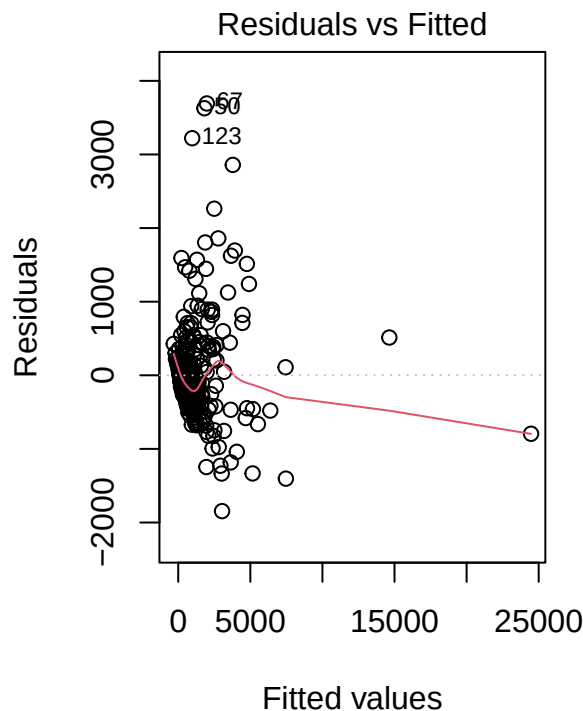
Conclusion The model is “ $Y = -61.57779 + 0.000798X_1 + 0.096053X_2 + -0.059023X_3 + 39.738897X_{4.1} + 26.382054X_{4.2} + 74.101108X_{4.3}$ ”, We conclude that potential problems with the regional dummies which are not significant. We keep them for specification, the adjusted R^2 is 90%

##c-) Test whether any geographic effects are present; use $\alpha = .10$. State the alternatives, decision rule, and conclusion. What is the P-value of the test? **Conclusion** Geographic effects are not significant at 90% confidence level. We build a reduced model excluding the geographic effects and we apply the Generalized Linear test. The $\alpha=0.01$ and the statistic the F_GLM , then null and alternative hypothesis as follows: Reject the null, geographic variables are not significant. Null and Alternative hypothesis Decision Rule: Reject H_0 in favor of H_a if p-value < 0.01 for the F_GLM .

##d-) Check the constant variance assumption for the errors. **Conclusion** One can check using plots and also a quantitative test like the Breush Pagan. When using the version of the Breush Pagan test used in the class materials it signal that heteroskedasticity is present (p-value 0.00 is lower than 0.05, thus, we reject the null hypothesis of homoscedasticity). Plots Residual vs. Fitted looks conclusive.

```
library(lmtest)
par(mfrow=c(1,2))
plot(model, which=1)
## Confirm using bptest() R function:
bptest(model, studentize = FALSE) # We have a relatively large number of observations n>100, so we select

##
## Breusch-Pagan test
##
## data: model
## BP = 161.3, df = 6, p-value < 0.00000000000000022
```



e-)Check the normality assumption.

ConclusionResiduals deviate from normality (according to the Normal Q-Q plot using confidence bands). Various residual normality tests confirm our observation. All quantitative tests, for example the Shapiro Wilks test have p-values < 0.05 , so we reject the null hypothesis of normality of the residuals.

```
library(olsrr)
ols_test_normality(model)
```

```
## -----
##          Test          Statistic      pvalue
## -----
## Shapiro-Wilk           0.7568        0.0000
## Kolmogorov-Smirnov      0.2009        0.0000
## Cramer-von Mises       48.2539        0.0000
## Anderson-Darling       30.1272        0.0000
## -----
```

```
# Load libraries again just in case
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr 2.1.5
## v lubridate 1.9.4    v stringr 1.5.1
## v purrr 1.0.4       v tibble 3.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x purrr::%@%()          masks rlang::%@%()
## x tinytex::check_installed() masks rlang::check_installed()
## x tidyr::expand()       masks reshape::expand()
## x dplyr::filter()       masks stats::filter()
## x purrr::flatten()      masks rlang::flatten()
## x purrr::flatten_chr()  masks rlang::flatten_chr()
## x purrr::flatten_dbl()  masks rlang::flatten_dbl()
## x purrr::flatten_int()  masks rlang::flatten_int()
## x purrr::flatten_lgl()  masks rlang::flatten_lgl()
## x purrr::flatten_raw()  masks rlang::flatten_raw()
## x purrr::invoke()       masks rlang::invoke()
## x dplyr::lag()          masks stats::lag()
## x purrr::lift()         masks caret::lift()
## x reshape::rename()     masks dplyr::rename()
## x dplyr::select()       masks MASS::select()
## x purrr::splice()       masks rlang::splice()
## x lubridate::stamp()    masks reshape::stamp()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidyr)
# Step 1 Rank Order your estimated residuals
ei<-model$residuals
ei.rank <- rank(ei)
## Step 3- Create some auxiliary variables n and p
n <- length(ei.rank)
p <- length(model$coefficients)

## Step 4 From 3.6 in ALSM page # 111
zr <- (ei.rank-0.375)/(n+0.25)
f.summary=summary(model)
```

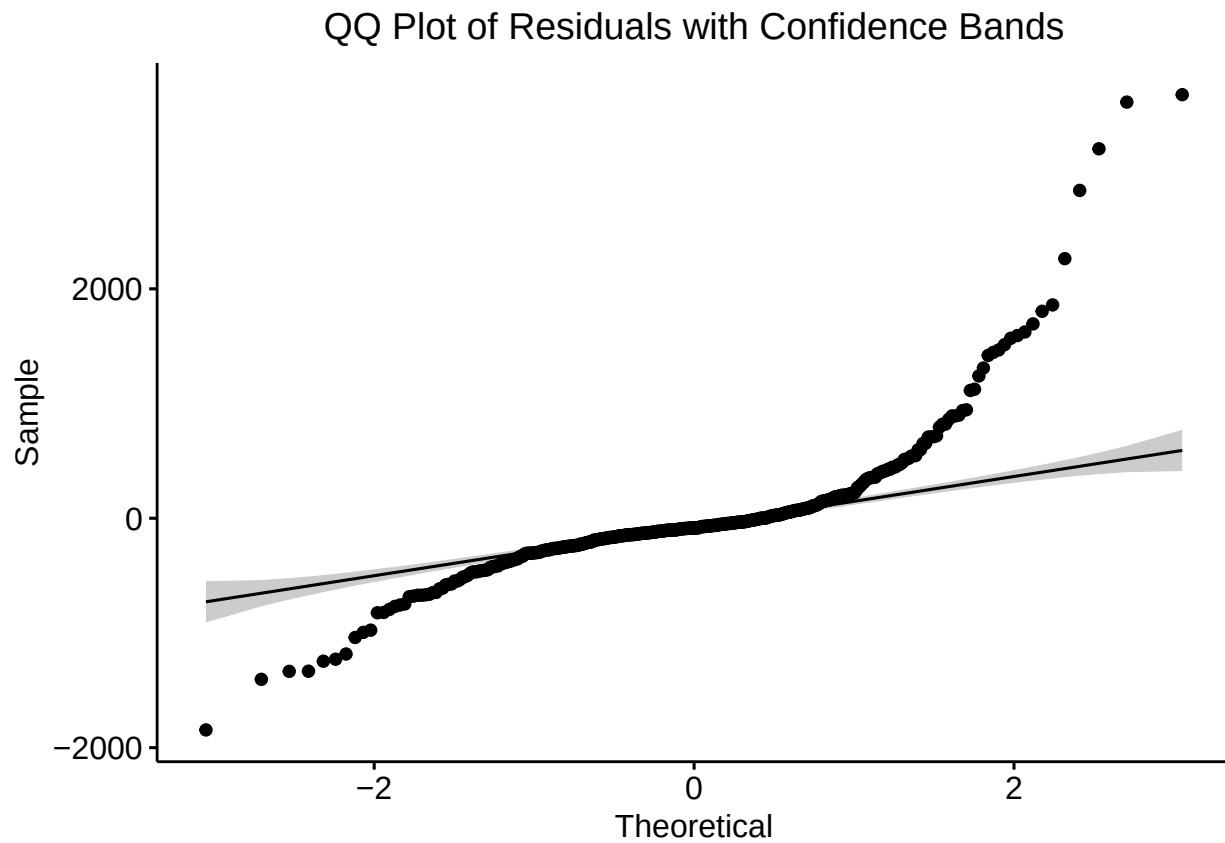
```

## Step 5 residual standard error from our summary above
zr1 = f.summary$sigma*qnorm(zr)
# Step 6 do a visual
library(ggplot2)
library(ggpubr)
a4<-anova(model)
mse<-a4$`Mean Sq`[4]
par(mfrow=c(2,2))
plot(zr1,ei,
xlab="Expected Value under Normality",
ylab="Residuals",
main="Residuals vs Expected Value under Normality")
#Residuals are normal but with one outlier point at the extreme, see below
ggqqplot(ei, title = "QQ Plot of Residuals with Confidence Bands") +
theme(plot.title = element_text(hjust = 0.5))

## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## i The deprecated feature was likely used in the ggpubr package.
## Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## i The deprecated feature was likely used in the ggpubr package.
## Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```



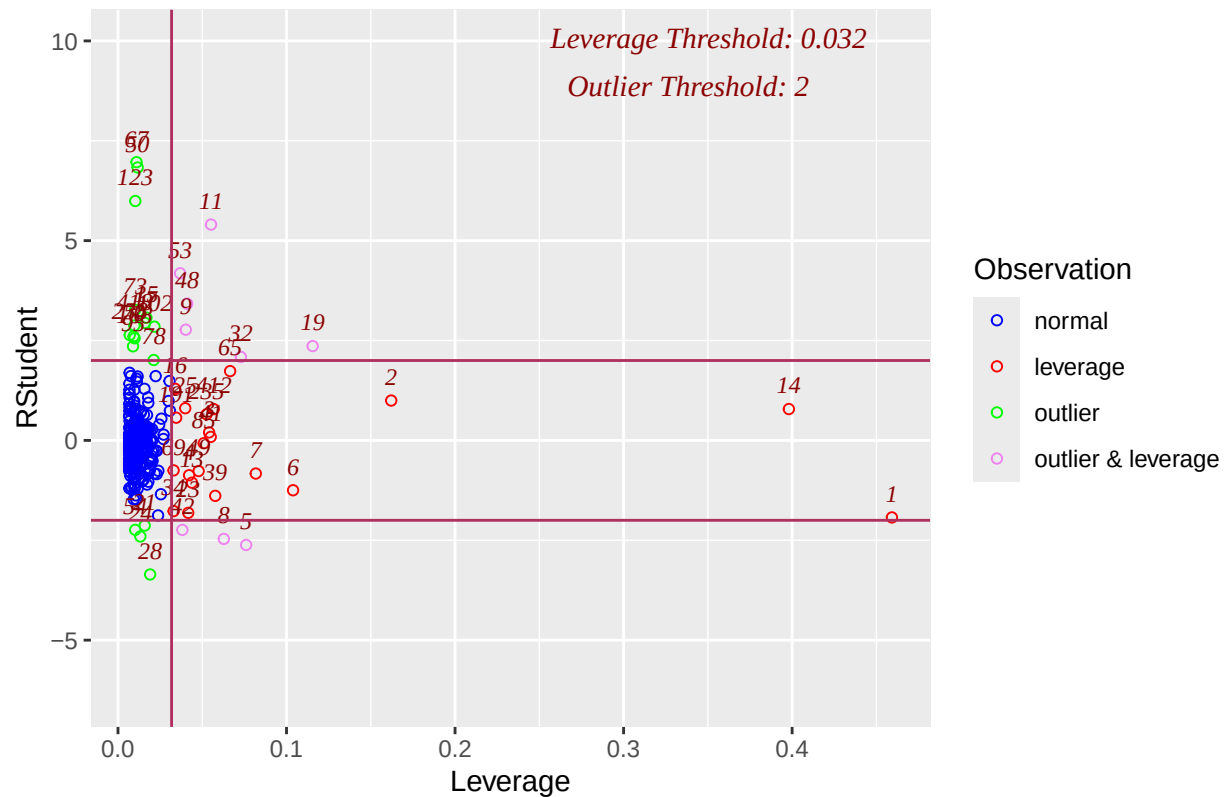
f-)Check for large leverage points.

ConclusionA simple way to check is to use a graph that way we have a visual that confirms multiple large leveraged points. Please refer to code below.

```
library(olsrr)

ols_plot_resid_lev(model)
```

Outlier and Leverage Diagnostics for Y



```
# Calculate Cook's distance
cooks_dist <- cooks.distance(model)

# Identify observations identified in the OLS PLOT RESID LEV graph
large_leverage_points <- which(cooks_dist > 3 / n) # You can adjust the threshold (e.g., 3/n) to match

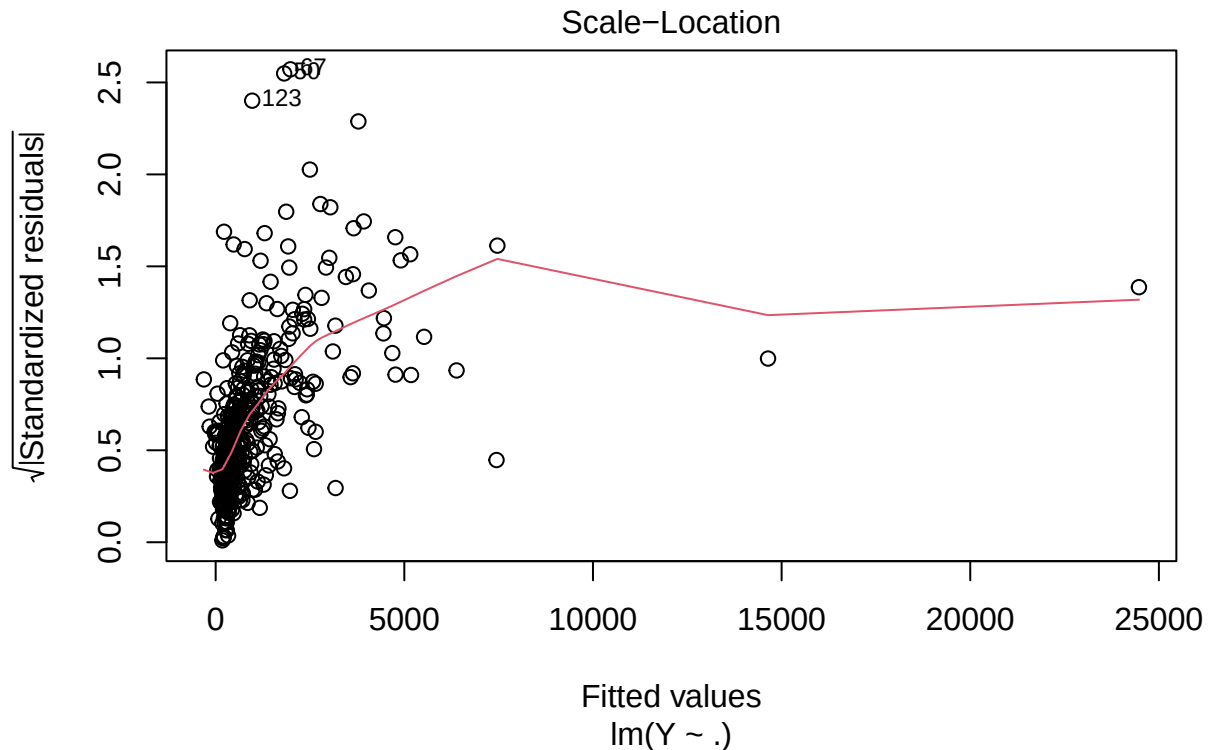
# Print the indices of observations above the desired threshold on the Cook's distance
print(large_leverage_points)
```

```
## 1 2 5 6 7 8 9 11 12 13 14 15 16 17 18 19 21 23 24 28
## 1 2 5 6 7 8 9 11 12 13 14 15 16 17 18 19 21 23 24 28
## 32 34 39 42 48 50 53 54 58 65 67 70 73 78 95 102 123 168 258 418
## 32 34 39 42 48 50 53 54 58 65 67 70 73 78 95 102 123 168 258 418
```

g-)Check for outliers.

Conclusion Many of the high Leverage points are also outliers. See code below.

```
plot(model,which=3)
```



```
# Identify observations identified in the OLS PLOT RESID LEV graph as outliers
outlier_points <- which(cooks_dist > 3 / n) # You can adjust the threshold (e.g., 3/n) to match the ou

# Print the indices of observations above the desired threshold on the Cook's distance
print(outlier_points)
```

```
## 1 2 5 6 7 8 9 11 12 13 14 15 16 17 18 19 21 23 24 28
## 1 2 5 6 7 8 9 11 12 13 14 15 16 17 18 19 21 23 24 28
## 32 34 39 42 48 50 53 54 58 65 67 70 73 78 95 102 123 168 258 418
## 32 34 39 42 48 50 53 54 58 65 67 70 73 78 95 102 123 168 258 418
```

h-)Check for influential points.

ConclusionSeveral influential points according Cook's distance. To decide if it is influential enough, I apply the Bonferroni test. The points 11, 50, 53, 67 and 123 are bad outliers according to the rule. See code below.

```
# Use a visual
```

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

```
## The following object is masked from 'package:purrr':
```

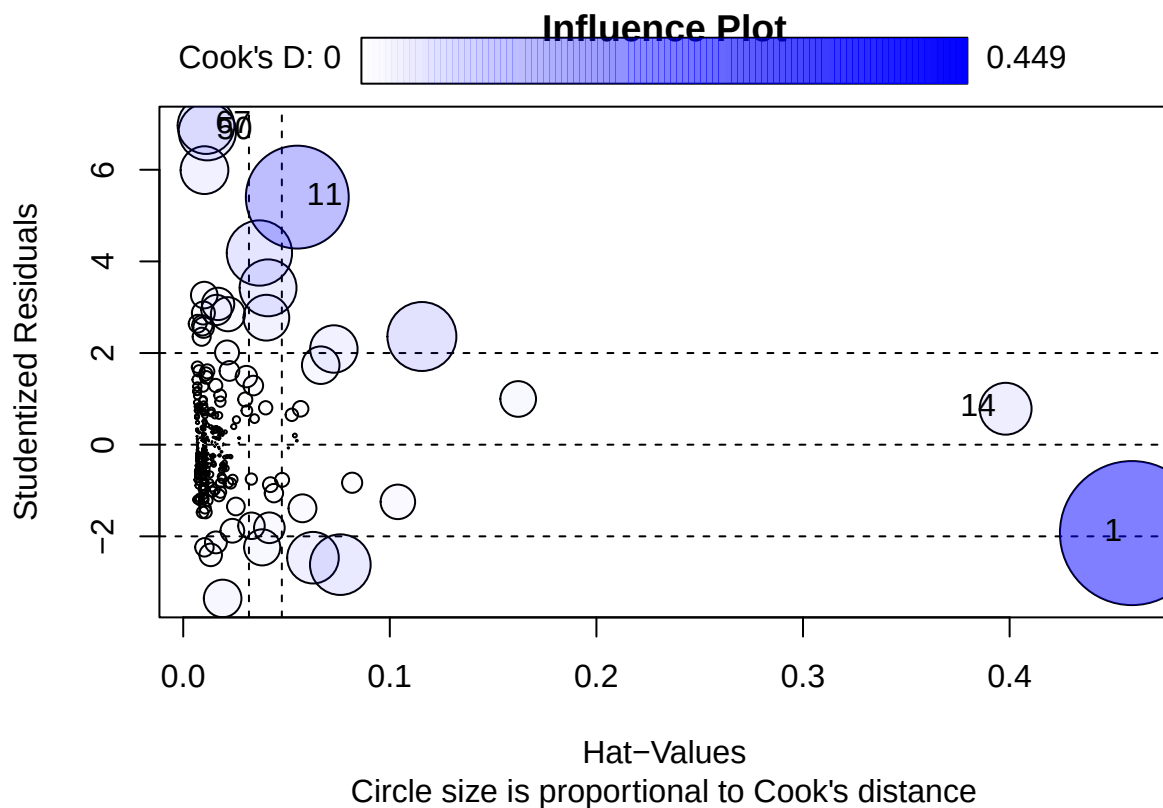
```
##
```

```
## some
```

```
## The following object is masked from 'package:dplyr':
##
##   recode
## The following objects are masked from 'package:faraway':
##
##   logit, vif
```

```
influencePlot(model, id="noteworthy", main="Influence Plot",
              sub="Circle size is proportional to Cook's distance")
```

```
## Warning in applyDefaults(id, defaults = list(method = "noteworthy", n = 2, :
## unnamed id arguments, will be ignored
```



```
##      StudRes      Hat      CookD
## 1  -1.9291424 0.45921003 0.44863388
## 11  5.4025571 0.05521312 0.22878071
## 14  0.7844021 0.39807193 0.05818117
## 50  6.8308615 0.01167584 0.07123639
## 67  6.9639907 0.01102595 0.06960599
```

```
#Compute Bonferroni Critical Value is 4.199773
Bonferroni_Critical_Value<-abs(qt(.05/(n*p),n-p-1))
```

```
#are there of them bad outliers?
outlier_test <- outlierTest(model)
```

```
knitr::kable(cbind(abs(outlier_test$rstudent), "<", Bonferroni_Critical_Value), align = "c",
```



```
caption = "Largest Outlier vs. Bonferroni Critical Value")
```

Table 1: Largest Outlier vs. Bonferroni Critical Value

			Bonferroni_Critical_Value
67	6.96399070263355	<	4.19977342859958
50	6.83086151621852	<	4.19977342859958
123	5.99151216343228	<	4.19977342859958
11	5.40255708956711	<	4.19977342859958
53	4.18292211164125	<	4.19977342859958

#Since Abs(-3.124428) is lower than 3.832128, we conclude that point West Virginia or Utah (2.52) are i
 outlier_test

```
##      rstudent unadjusted p-value Bonferroni p
## 67  6.963991  0.000000000012380 0.000000005447
## 50  6.830862  0.000000000028785 0.000000012665
## 123 5.991512  0.000000004385000 0.000001929400
## 11  5.402557  0.000000108760000 0.000047856000
## 53  4.182922  0.000034869000000 0.015343000000
```

Problem 2

Using the prostate data under faraway library, if you are using the R Studio, please use the command below to download the data:

```
library(faraway)
data("prostate", package="faraway")
```

If you are using Posit R Cloud instead of RStudio Desktop, please use the command below to download the data:

```
tmp <- tempfile(fileext = ".rda")
download.file(
  "https://raw.githubusercontent.com/cran/faraway/master/data/prostate.rda",
  destfile = tmp,
  mode = "wb"
)
load(tmp)
head(prostate)
```

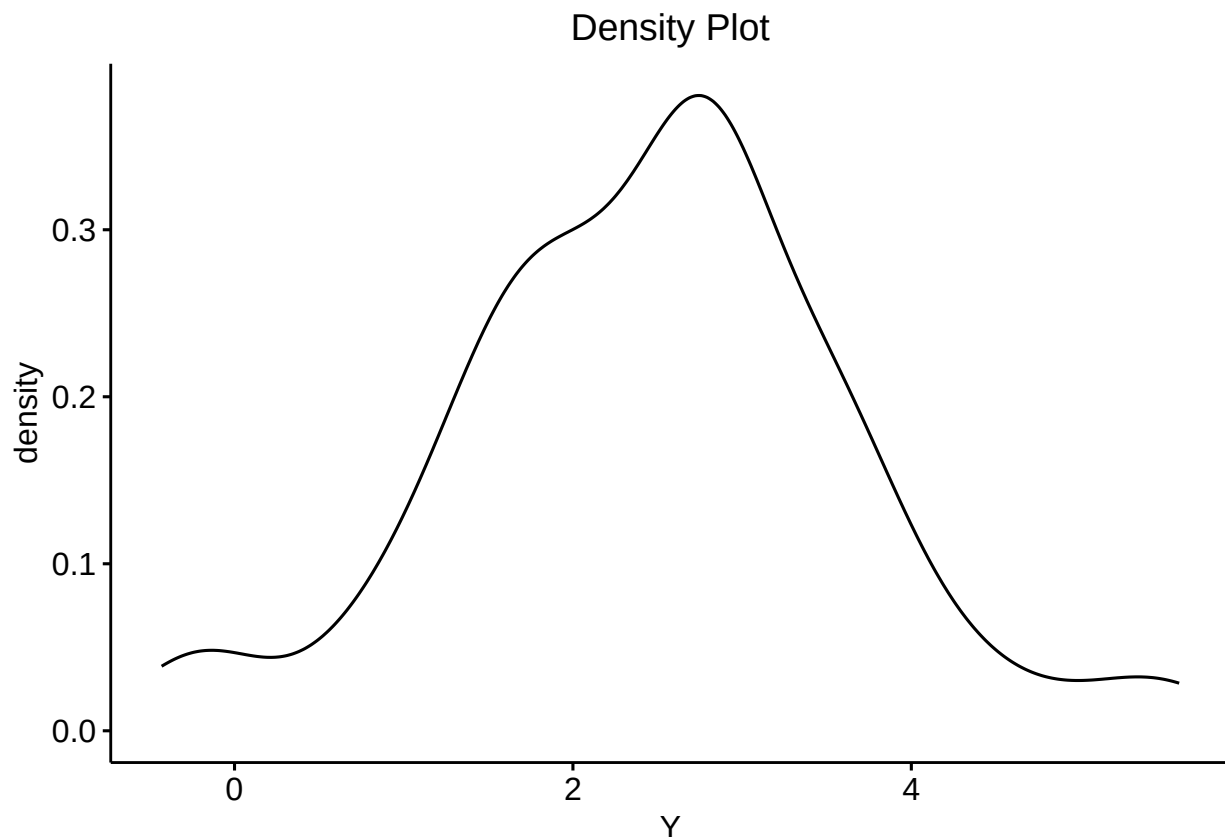
```
##      lcavol lweight age      lbph svi      lcp gleason pgg45      lpsa
## 1 -0.5798185 2.7695 50 -1.386294 0 -1.38629      6      0 -0.43078
## 2 -0.9942523 3.3196 58 -1.386294 0 -1.38629      6      0 -0.16252
## 3 -0.5108256 2.6912 74 -1.386294 0 -1.38629      7     20 -0.16252
## 4 -1.2039728 3.2828 58 -1.386294 0 -1.38629      6      0 -0.16252
## 5  0.7514161 3.4324 62 -1.386294 0 -1.38629      6      0  0.37156
## 6 -1.0498221 3.2288 50 -1.386294 0 -1.38629      6      0  0.76547
```

Use lpsa as the response and all the other variables as predictors to answer the questions below. (30 points)

##a-) Create train and test data sets. Use 70% of the data for train data set and use remaining data (30% of data) for test data set (use set.seed(1023)). Build a regression model to predict lpsa on the train data set. Write down the regression model and comment on the model performance. (5 points)

CONCLUSION The regression model is “ $Y = 1.063527 + 0.699881lcavol + 0.395885lweight + -0.029109age + 0.111465lbph + 0.663679svi + -0.257964lcp + 0.064838gleason + 0.007046pgg45$ ”, the graphs show the residuals vs. fitted shows that the errors seem randomly distributed with constant variance with evidence of linearity as indicated by the red line. The normal QQ plot does not show evidence of fat tails although borderline. And a few outliers. However the overall fit indicate most of the predictors are not significant using alpha of 0.05 with an adjusted R2 of 59%. (we keep them for further analysis) See code below.

```
library(faraway)
library(caret)
# Split the data into training and testing samples
set.seed(1023)
#Are we modeling a stationary variable (distributed with constant mean and variance) or a stochastic, r
ggdensity(prostate$lpsa,
title = "Density Plot",
xlab = "Y") +
theme(plot.title = element_text(hjust = 0.5))
```



```
#split the data into two parts 70-30 split
DataSplit<-createDataPartition(y = prostate$lpsa, p = 0.69, list = FALSE)
#creating the train and test data set
prostate.train<-prostate[DataSplit,]
prostate.test<-prostate[-DataSplit,]
# always check if you have 70%
num_observations <- nrow(prostate)
num_observations_train <- nrow(prostate.train)
size_test <- (num_observations_train / num_observations)*100
```

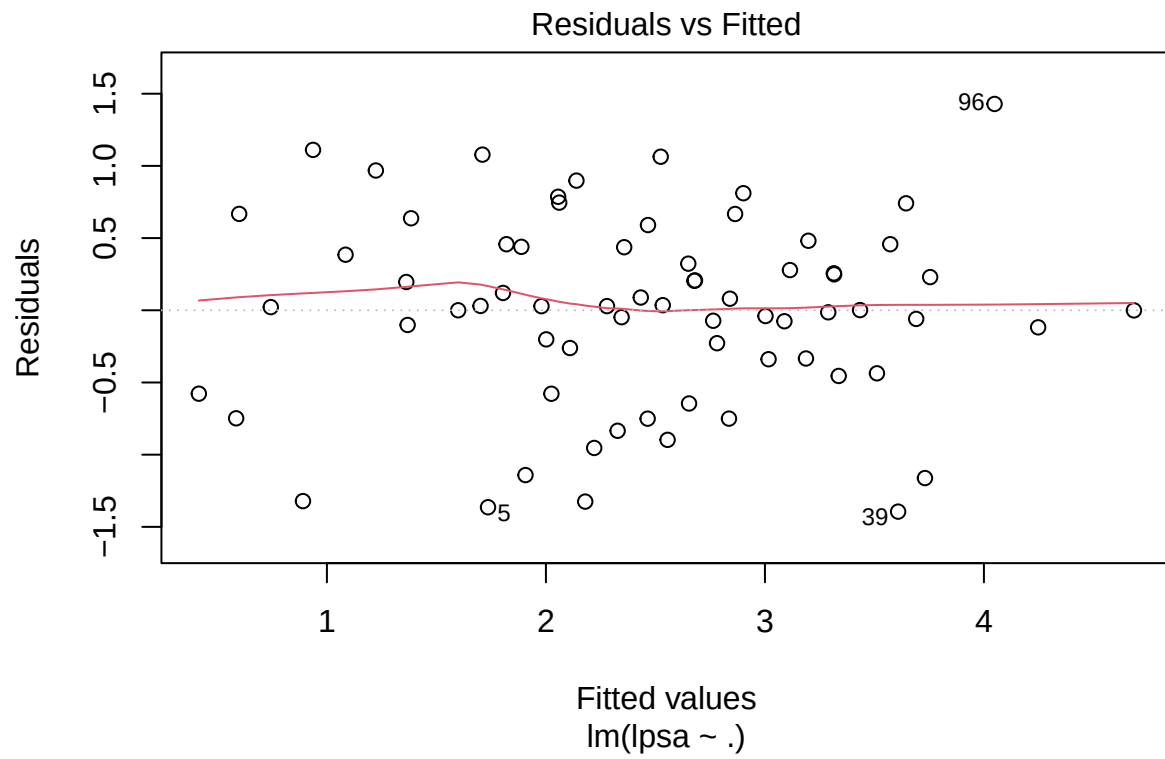
```
# we conclude 71% is sufficiently close to 70%, note the R-Developer may have considered other objectives
model <- lm(lpsa ~ ., data = prostate.train)
summary(model)
```

```
##
## Call:
## lm(formula = lpsa ~ ., data = prostate.train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.39448 -0.33906  0.02852  0.43937  1.42896
##
## Coefficients:
##              Estimate Std. Error t value    Pr(>|t|)
## (Intercept)  1.063527   1.555514   0.684    0.4968
## lcavol       0.699881   0.107840   6.490 0.0000000184 ***
## lweight      0.395885   0.191228   2.070   0.0427 *
## age         -0.029109   0.013258  -2.196   0.0320 *
## lbph        0.111465   0.070438   1.582   0.1188
## svi         0.663679   0.283099   2.344   0.0224 *
## lcp        -0.257964   0.120597  -2.139   0.0365 *
## gleason     0.064838   0.190347   0.341   0.7346
## pgg45       0.007046   0.005962   1.182   0.2419
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6953 on 60 degrees of freedom
## Multiple R-squared:  0.6706, Adjusted R-squared:  0.6267
## F-statistic: 15.27 on 8 and 60 DF,  p-value: 0.000000000005868
```

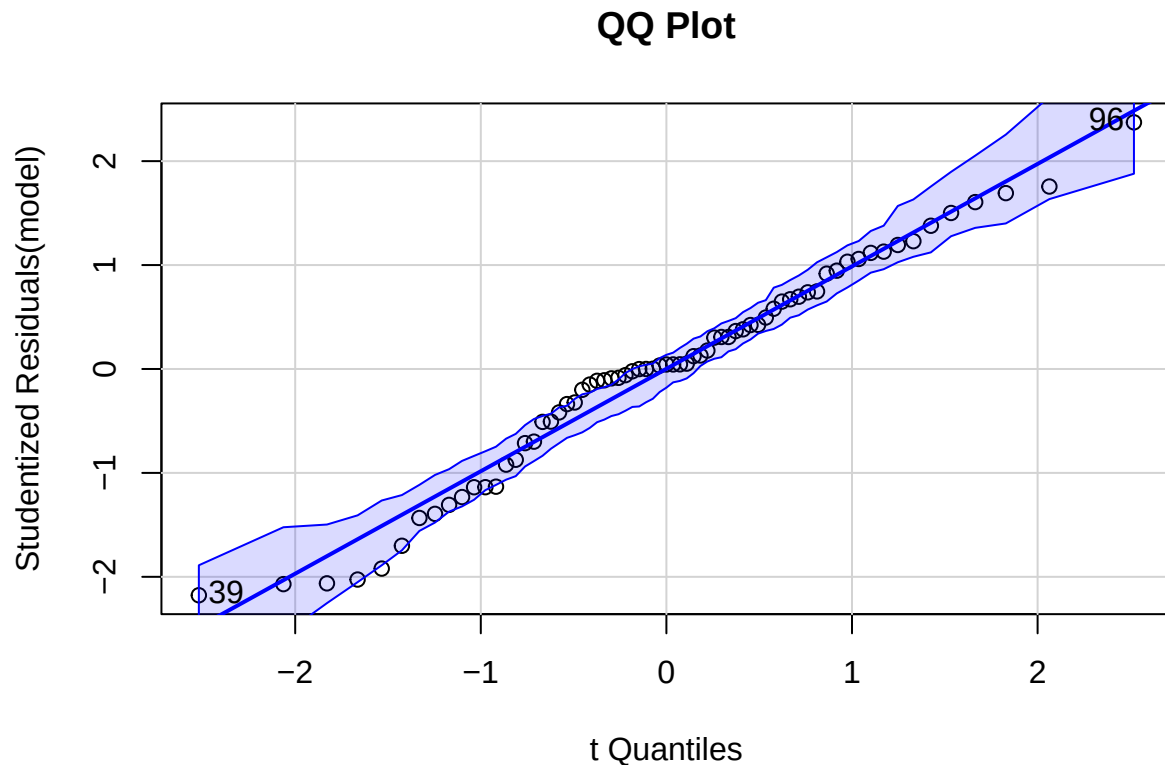
```
dispRegFunc(model)
```

```
## [1] "Y = 1.063527 + 0.699881lcavol + 0.395885lweight + -0.029109age + 0.111465lbph + 0.663679svi + -0.257964lcp + 0.064838gleason + 0.007046pgg45"
```

```
plot(model, which=1)
```



```
qqPlot(model, main = "QQ Plot")
```



```
## 39 96
```

```
## 28 69
```

```
print(paste("Percentage in Train: ",size_test))
```

```
## [1] "Percentage in Train: 71.1340206185567"
```

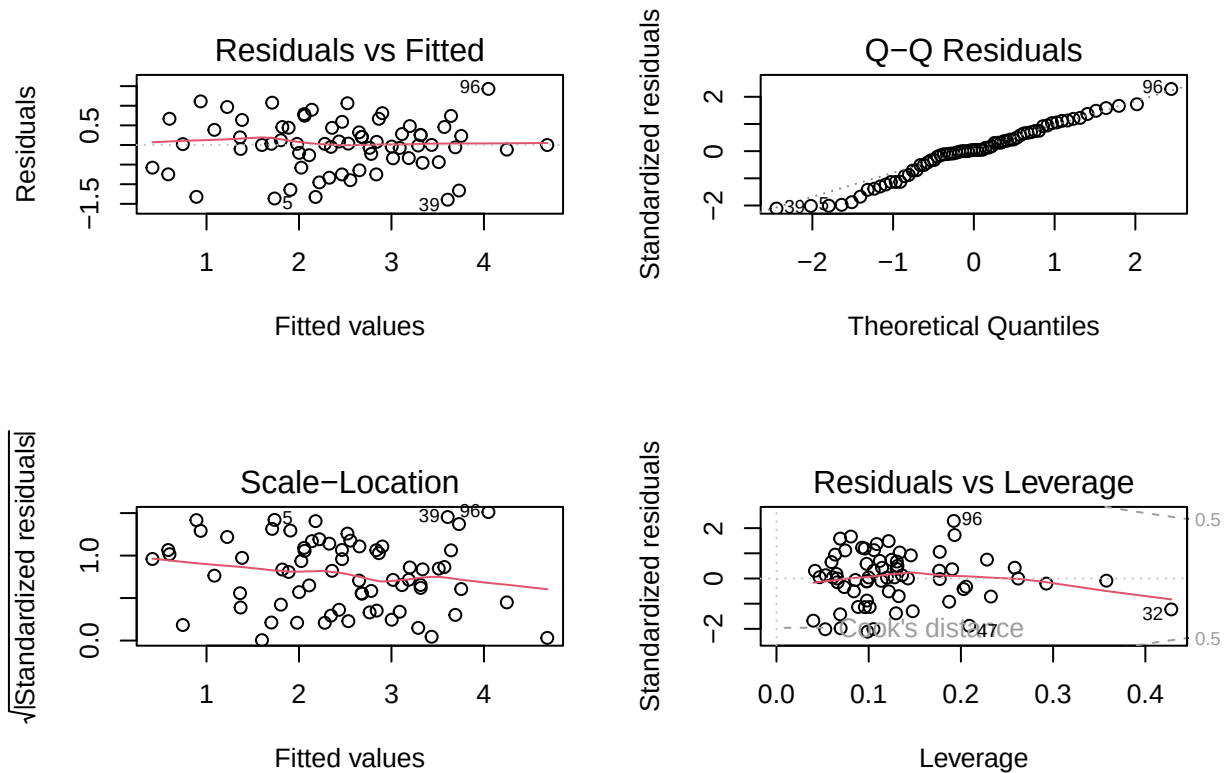
```
dispRegFunc(model)
```

```
## [1] "Y = 1.063527 + 0.699881lcavol + 0.395885lweight + -0.029109age + 0.111465lbph + 0.663679svi + -"
```

##b-) Visually by using the graphs, comments on regression model assumptions. **CONCLUSION** The regression model for analysis is “ $Y = 1.063527 + 0.699881lcavol + 0.395885lweight + -0.029109age + 0.111465lbph + 0.663679svi + -0.257964lcp + 0.064838gleason + 0.007046pgg45$ ”, below,graph shows residual distribution with no heavy tails. There is some outliers, but most of the assumption seem to be in line. See code below.

Analysis **A1) Is normal linear regression model is appropriate for the problem? 1.-Linearity of Regression Function: The Residual Vs. Fitted plot show a linear relationship, confirming the OLS is appropriate. 2.-Error Variance seems constant. The pattern of the Residual vs. Fitted also seems to confirm that the variance is random. 3.-Presence of outliers. The Residual vs. Leverage plot show there is at least one large no outliers. A2) Are the estimated errors (residuals) are independent from the Xs and from the estimated values $\hat{Y}$? 4.-Non-independence of Error Terms. Residual vs. Fitted and the Scale Location plots show strong no correlations among residuals. However, predictors are not significant with a relatively high R^2 that may point to collinearity or lack of independence of the predictors. 5.-Omission of Important Predictor or explanatory variables. Residuals pattern seem to be random there is no systematic influence of the omitted variable here. *A3) Are the errors are iid- $N(0, \sigma)$? that is the errors are independent and identically normally distributed. 6. Normality of the Residuals. The QQ plot shows some deviation from the diagonal at the middle, pointing to some non-normalities.

```
par(mfrow=c(2,2))
plot(model)
```



```
dispRegFunc(model)
```

```
## [1] "Y = 1.063527 + 0.699881lcavol + 0.395885lweight + -0.029109age + 0.111465lbph + 0.663679svi + -0.257964lcp"
# Prepare a scatter plot of the data
summary(model)
```

```
##
## Call:
## lm(formula = lpsa ~ ., data = prostate.train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.39448 -0.33906  0.02852  0.43937  1.42896
##
## Coefficients:
##              Estimate Std. Error t value    Pr(>|t|)
## (Intercept)  1.063527   1.555514   0.684    0.4968
## lcavol       0.699881   0.107840   6.490 0.0000000184 ***
## lweight      0.395885   0.191228   2.070   0.0427 *
## age        -0.029109   0.013258  -2.196   0.0320 *
## lbph        0.111465   0.070438   1.582   0.1188
## svi         0.663679   0.283099   2.344   0.0224 *
## lcp        -0.257964   0.120597  -2.139   0.0365 *
```

```
## gleason      0.064838  0.190347  0.341      0.7346
## pgg45        0.007046  0.005962  1.182      0.2419
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6953 on 60 degrees of freedom
## Multiple R-squared:  0.6706, Adjusted R-squared:  0.6267
## F-statistic: 15.27 on 8 and 60 DF,  p-value: 0.00000000005868
```

##c-) Test the model performance on the test data and comment of model stability and performance.
****Conclusion:** Model seem unstable with a 10% variation between train and test dataset. We confirm the fact that linear fit is overfit because the model performance is worst in the test data. Model may be improved by excluding not-significant variables.

```
library(caret)
PredictedTrain<-predict(model,prostate.train)
ModelTrain<-data.frame(obs = prostate.train$lpsa, pred=PredictedTrain)
Model_Train=defaultSummary(ModelTrain)

PredictedTest<-predict(model,prostate.test)
ModelTest<-data.frame(obs = prostate.test$lpsa, pred=PredictedTest)
Model_Test=defaultSummary(ModelTest)

g = rbind(Model_Train,Model_Test)
g
```

```
##              RMSE  Rsquared      MAE
## Model_Train 0.6483976 0.6706072 0.4994026
## Model_Test  0.8086777 0.5853874 0.5998993
```

##d-) Perform Breusch-Pagan Test, write down null and alternative hypotheses and your conclusion.
CONCLUSION We reject the null hypothesis of constant variance if the value of our test statistics is below a critical value, or equivalently if the p.value associated with our test statistics is well above $\alpha = .05$. Here we can not reject the null hypothesis of constant variance.

#We reject the null hypothesis as p.value is 0.6812 > 0.05 - so, we conclude that the error variances are not constant

```
library(lmtest)
bptest(model)
```

```
##
## studentized Breusch-Pagan test
##
## data:  model
## BP = 5.6964, df = 8, p-value = 0.6812
```

e-) Use the Box-Cox method to determine the best transformation on the response. if transformation is needed, perform the transformation and rebuild the model on the train data and check the model stability on the test data set. (10 points) **CONCLUSION** Box-Cox can not be used, lpsa is not a positive variable in all its range. We propose instead to drop non-significant variables to rebuild the model and test for stability. Seems overfit as the resulting model is better in sample. See code below.

```
# 1. Update to drop not significant variables
model2<- lm(lpsa ~ ., data = prostate.train[,c(-3,-4,-5,-6,-7,-8)])
summary(model2)

##
## Call:
## lm(formula = lpsa ~ ., data = prostate.train[, c(-3, -4, -5,
```

```
##      -6, -7, -8)])
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -1.61306 -0.39664  0.06953  0.55399  1.90776
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)   0.1092     0.6028   0.181      0.8568
## lcavol        0.7007     0.0799   8.769 0.00000000000112 ***
## lweight       0.3818     0.1676   2.278     0.0259 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.727 on 66 degrees of freedom
## Multiple R-squared:  0.6039, Adjusted R-squared:  0.5919
## F-statistic: 50.32 on 2 and 66 DF,  p-value: 0.0000000000005326
```

```
# 2. Performance new model out of sample
library(caret)
PredictedTrain<-predict(model2,prostate.train)
ModelTrain<-data.frame(obs = prostate.train$lpsa, pred=PredictedTrain)
Model_Train=defaultSummary(ModelTrain)

PredictedTest<-predict(model2,prostate.test)
ModelTest<-data.frame(obs = prostate.test$lpsa, pred=PredictedTest)
Model_Test=defaultSummary(ModelTest)

g = rbind(Model_Train,Model_Test)
g
```

```
##              RMSE Rsquared      MAE
## Model_Train 0.7109998 0.6039315 0.5565267
## Model_Test  0.8143091 0.5392787 0.6623440
```

Problem 3

Commercial properties data set. A commercial real estate company evaluates vacancy rates, square footage, rental rates, and operating expenses for commercial properties in a large metropolitan area in order to provide clients with quantitative information upon which to make rental decisions. The data are taken from 81 suburban commercial properties include the age (X1), operating expenses and taxes (X2), vacancy rates (X3), total square footage (X4), and rental rates (Y). (30 points) ### Data Preparations, upload data, libraries, install GGally and corrrplot first then you commented out otherwise PDF is not going to work.

##a-) Fit regression model for four predictor variables to the data. State the estimated regression function.(5 points)

Conclusion The fitted regression is “ $Y = 12.200586 - 0.142034X1 + 0.282017X2 + 0.619344X3 + 0.0000079243X4$ ”. And by looking at the summary of regression model we see that X1, X2, X4 have p value less than 0.05 and all three variables are significant, on other hand X3 is not significant. Model adjusted R-squared 56%. Model p value is p-value: 0, so model is also significant.

```
options("scipen" = 12) ##Used to avoid displaying numbers such as 1+e3 (exponent), changing the option
f1<-lm(Y~X1+X2+X3+X4, data= d1)
summary(f1)
```

```
##
```



```
## Call:
## lm(formula = Y ~ X1 + X2 + X3 + X4, data = d1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.1872 -0.5911 -0.0910  0.5579  2.9441
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 12.20058582  0.577956174  21.110    < 2e-16 ***
## X1          -0.142033644  0.021342610  -6.655 0.00000000389 ***
## X2           0.282016530  0.063172350   4.464 0.00002747396 ***
## X3           0.619343503  1.086812829   0.570      0.57
## X4           0.000007924  0.000001385   5.722 0.00000019760 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.137 on 76 degrees of freedom
## Multiple R-squared:  0.5847, Adjusted R-squared:  0.5629
## F-statistic: 26.76 on 4 and 76 DF,  p-value: 0.00000000000007272
```

```
dispRegFunc(f1)
```

```
## [1] "Y = 12.200586 + -0.142034X1 + 0.282017X2 + 0.619344X3 + 0.000008X4"
```

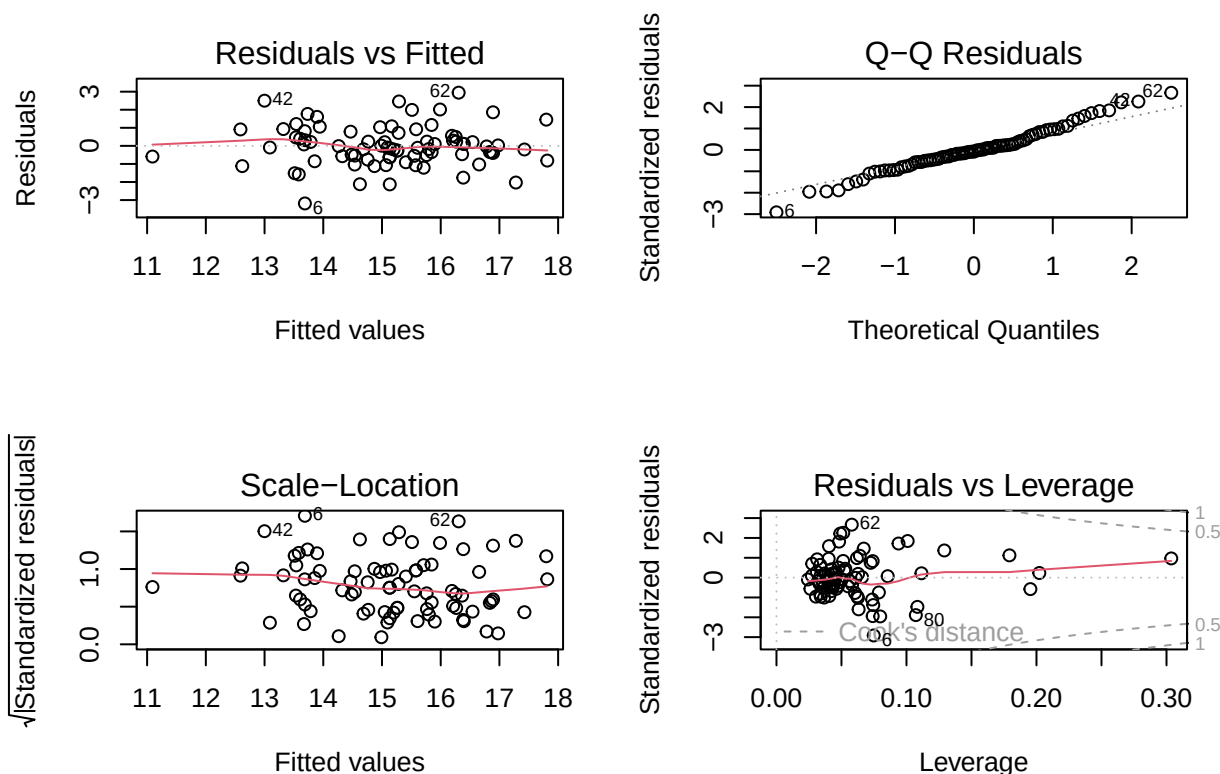
##b-) Visually using the appropriate graphs, comment on the regression model assumptions. (5 points)
CONCLUSION The regression model is “ $Y = 12.200586 + -0.142034X1 + 0.282017X2 + 0.619344X3 + 0.000008X4$ ”, graph shows that the residual distribute approximately normal without heavy tails. The variance seems constant with constant mean and the overall fit linear.

Analysis **A1) Is normal linear regression model is appropriate for the problem? 1.-Linearity of Regression Function: The Residual Vs. Fitted plot shows a linear relationship. 2.-Constancy of the Error Variance. The pattern of the Residual vs. Fitted also seems to confirm that the variance is constant with homogeneous dispersion. 3.-Presence of outliers. The Residual vs. Leverage plot show there is no outlier points. A2) Are the estimated errors (residuals) are independent from the Xs and from the estimated values $\hat{Y}$? 4.-Non-independence of Error Terms. Residual vs. Fitted and the Scale Location plots show random dispersion of residuals. Predictor variables seem independent. 5.-Omission of Important Predictor or explanatory variables. Some graphs show to us that there is no omitted variable as disclosed by the residuals pattern, it is random there is no systematic influence of the omitted variable here. *A3) Are the errors are iid~N(0,sigma)?? Yes errors seem independent and identically normally distributed. 6. Non-normality of the Residuals. The QQ plot with confidence bands does not shows large deviations from the diagonal at the extremes, pointed to normality.

```
library(car)
qqPlot(f1, main = "QQ Plot")
```



```
## [1] 6 62  
par(mfrow=c(2,2))  
plot(f1)
```



c-) Divide the 81 cases into two groups. placing the 40 cases with the smallest fitted values into group 1 and the remaining cases into group 2. Conduct the Brown-Forsythe test.(5 points)

Conclusion We test homogeneity of error variances using the Brown-Forsythe test (Levene's test with the median) applied to the model residuals. Following the instructions, we split the data into two groups: the 40 observations with the smallest **fitted values** and the remaining observations. The null hypothesis is equal residual variances across these two groups. The test returns $p > 0.05$, so we fail to reject H_0 ; there is no evidence of heteroscedasticity in the residuals.

```
# ---- 1(e) Brown-Forsythe test exactly per spec ----
library(car)

# Residuals and fitted from the Problem 1 model
r      <- residuals(f1)
fhat   <- fitted(f1)

# Group by the 40 smallest fitted values (group A) vs the rest (group B)
ord <- order(fhat)
grp <- rep("B_rest", length(fhat))
grp[ord[1:40]] <- "A_small40"
grp <- factor(grp, levels = c("A_small40", "B_rest"))

# Brown-Forsythe test (Levene with center = "median") on residuals by group
bf <- leveneTest(r ~ grp, center = "median")
print(bf)
```

```
## Levene's Test for Homogeneity of Variance (center = "median")
##      Df F value Pr(>F)
```

```
## group 1 0.3048 0.5824
##      79
```

##d-) Fit a second order model ($Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4 + b_5X_1^2 + b_6X_2^2 + b_7X_3^2 + b_8X_4^2 + b_9X_1X_2 + b_{10}X_1X_3 + b_{11}X_1X_4 + b_{12}X_2X_3 + b_{13}X_2X_4 + b_{14}X_3X_4 + b_{15}X_1X_2X_3 + b_{16}X_1X_2X_4 + b_{17}X_2X_3X_4 + b_{18}X_1X_2X_3X_4 + b_{19}X_1X_3X_4$) Drop the insignificant variables from the model one at a time, refit the model and repeat this until all the variables you have in the model are significant. Compare this again the model in part b) (15 points) **Conclusion** See code below. The full model is $Y = 14.769155 - 0.21454x1g + 0.222764x2g - 6.30191x3g + 0.00000409571x4g + 0.00250229I(x1g^2) - 0.0327956I(x2g^2) - 9.83949I(x3g^2) - 0.000000000066436I(x4g^2) + 0.0254521x1g:x2g - 0.677002x1g:x3g - 0.000000458195x1g:x4g - 1.03039x2g:x3g + 0.00000159848x2g:x4g - 0.000019107x3g:x4g - 0.0590515x1g:x2g:x3g + 0.000000623596x1g:x2g:x4g + 0.0000077624x2g:x3g:x4g + 0.0000010229x1g:x3g:x4g + 0.00000488099x1g:x2g:x3g:x4g$. Adjusted R-squared is approximately 0.68; not all variables are significant at $\alpha = 0.05$. We apply backward elimination by p-value to remove non-significant terms and then confirm with a general linear test (reduced FIRST, full SECOND). Since the GLT p-value is $0.645 > 0.05$, the reduced model is acceptable and the dropped terms can be omitted.

```
# Centering the variables
x1g <- d1$X1 - mean(d1$X1)
x2g <- d1$X2 - mean(d1$X2)
x3g <- d1$X3 - mean(d1$X3)
x4g <- d1$X4 - mean(d1$X4)

# Full 2nd-order model (centered)
modell1 <- lm(
  Y ~ x1g + x2g + x3g + x4g +
    I(x1g^2) + I(x2g^2) + I(x3g^2) + I(x4g^2) +
    x1g:x2g + x1g:x3g + x1g:x4g + x2g:x3g + x2g:x4g + x3g:x4g +
    x1g:x2g:x3g + x1g:x2g:x4g + x2g:x3g:x4g + x1g:x3g:x4g +
    x1g:x2g:x3g:x4g,
  data = d1
)

summary(modell1)
```

```
##
## Call:
## lm(formula = Y ~ x1g + x2g + x3g + x4g + I(x1g^2) + I(x2g^2) +
##      I(x3g^2) + I(x4g^2) + x1g:x2g + x1g:x3g + x1g:x4g + x2g:x3g +
##      x2g:x4g + x3g:x4g + x1g:x2g:x3g + x1g:x2g:x4g + x2g:x3g:x4g +
##      x1g:x3g:x4g + x1g:x2g:x3g:x4g, data = d1)
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -2.01237 -0.60959 -0.02205  0.50532  1.96870
##
```

```
## Coefficients:
```

```
##              Estimate      Std. Error t value      Pr(>|t|)
## (Intercept)  14.769155190479328  0.395778488537179   37.317    < 2e-16 ***
## x1g          -0.214539628850376  0.030264297927062   -7.089 0.00000000163 ***
## x2g           0.222764123117447  0.111106868185483    2.005  0.049411 *
## x3g          -6.301910632993674  2.669569053738918   -2.361  0.021455 *
## x4g           0.000004095706597  0.000002090613639    1.959  0.054677 .
## I(x1g^2)      0.002502292098629  0.006782217648930    0.369  0.713443
## I(x2g^2)     -0.032795610566309  0.026579923034863   -1.234  0.221992
```

```
## I(x3g^2)          -9.839487814746784 11.191504867204950 -0.879      0.382747
## I(x4g^2)          -0.000000000006644 0.000000000013876 -0.479      0.633799
## x1g:x2g           0.025452133675648 0.016505894182542 1.542      0.128245
## x1g:x3g          -0.677001967220541 0.381190921242822 -1.776      0.080717
## x1g:x4g          -0.000000458194693 0.000000344753333 -1.329      0.188781
## x2g:x3g          -1.030392945790951 1.575322524502361 -0.654      0.515516
## x2g:x4g           0.000001598481328 0.000001171531273 1.364      0.177443
## x3g:x4g          -0.000019107048796 0.000030549989717 -0.625      0.534018
## x1g:x2g:x3g      -0.059051537950291 0.216780703777843 -0.272      0.786233
## x1g:x2g:x4g       0.000000623595572 0.000000169926269 3.670      0.000512 ***
## x2g:x3g:x4g       0.000007762395009 0.000015107243037 0.514      0.609235
## x1g:x3g:x4g       0.000001022901828 0.000004618531784 0.221      0.825460
## x1g:x2g:x3g:x4g  0.000004880990459 0.000002374884223 2.055      0.044143 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9804 on 61 degrees of freedom
## Multiple R-squared:  0.7521, Adjusted R-squared:  0.6749
## F-statistic: 9.743 on 19 and 61 DF, p-value: 0.00000000003775
```

```
dispRegFunc(model1)
```

```
## [1] "Y = 14.769155 + -0.21454x1g + 0.222764x2g + -6.301911x3g + 0.000004x4g + 0.002502I(x1g^2) + -0.
```

Model Selection by Stepwise Elimination, removing the largest p-value terms iteratively until all remaining terms are significant at 5%.

Here we want only significant variables at 5%

```
# Final removal to ensure all remaining terms are significant at 5%
model2 <- lm(Y ~ x1g + x2g + x3g + x4g +
             x1g:x2g + x1g:x4g + x1g:x2g:x4g + x1g:x2g:x3g:x4g,
             data = d1)
summary(model2)
```

```
##
## Call:
## lm(formula = Y ~ x1g + x2g + x3g + x4g + x1g:x2g + x1g:x4g +
##      x1g:x2g:x4g + x1g:x2g:x3g:x4g, data = d1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.25474 -0.61883 -0.00264  0.53946  2.37729
##
## Coefficients:
##              Estimate      Std. Error t value      Pr(>|t|)
## (Intercept)  14.79888222203  0.13044500540 113.449    < 2e-16 ***
## x1g          -0.18707502205  0.02082656829  -8.983 0.000000000000225 ***
## x2g           0.31872560601  0.05732785448   5.560 0.000000432162281 ***
## x3g          -3.62199953549  1.31472643462  -2.755   0.007429 **
## x4g           0.00000500879  0.00000139394   3.593   0.000594 ***
## x1g:x2g       0.03940190662  0.00970310653   4.061   0.000123 ***
## x1g:x4g      -0.00000056031  0.00000022313  -2.511   0.014282 *
## x1g:x2g:x4g   0.00000044645  0.00000008903   5.015 0.000003685336241 ***
## x1g:x2g:x3g:x4g 0.00000153165  0.00000075044   2.041   0.044918 *
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9649 on 72 degrees of freedom
## Multiple R-squared:  0.7166, Adjusted R-squared:  0.6852
## F-statistic: 22.76 on 8 and 72 DF,  p-value: < 0.00000000000000022

dispRegFunc(model2)

## [1] "Y = 14.798882 + -0.187075x1g + 0.318726x2g + -3.622x3g + 0.000005x4g + 0.039402x1g:x2g + -0.000005x1g:x3g + 0.000005x1g:x4g + 0.000005x2g:x3g + 0.000005x2g:x4g + 0.000005x3g:x4g"
# General Linear Test: reduced FIRST, full SECOND
anova(model2, model1)

## Analysis of Variance Table
##
## Model 1: Y ~ x1g + x2g + x3g + x4g + x1g:x2g + x1g:x4g + x1g:x2g:x4g +
##      x1g:x2g:x3g:x4g
## Model 2: Y ~ x1g + x2g + x3g + x4g + I(x1g^2) + I(x2g^2) + I(x3g^2) +
##      I(x4g^2) + x1g:x2g + x1g:x3g + x1g:x4g + x2g:x3g + x2g:x4g +
##      x3g:x4g + x1g:x2g:x3g + x1g:x2g:x4g + x2g:x3g:x4g + x1g:x3g:x4g +
##      x1g:x2g:x3g:x4g
##      Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1         72 67.029
## 2         61 58.632 11      8.397 0.7942 0.645
```